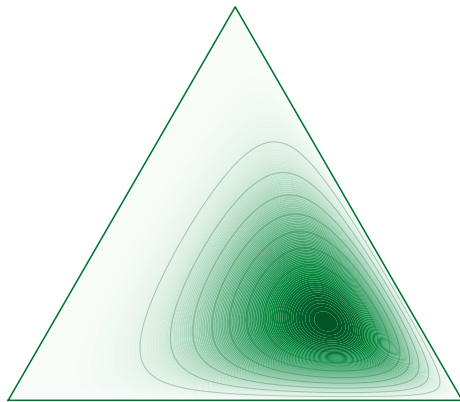


# Evidential Bayesian Neural Networks for Uncertainty-Aware Classification

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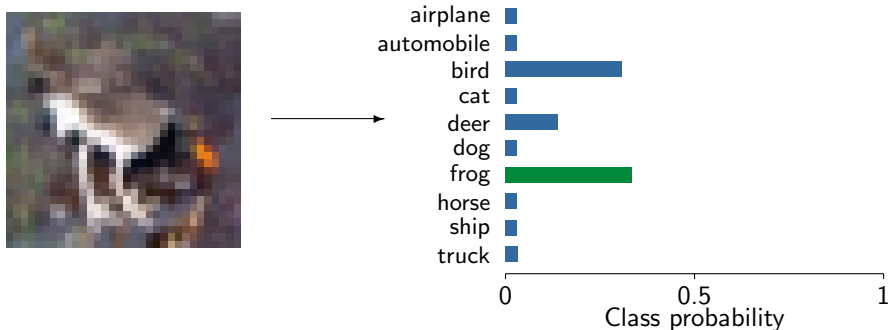
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# Why Uncertainty-Aware Modelling Matters

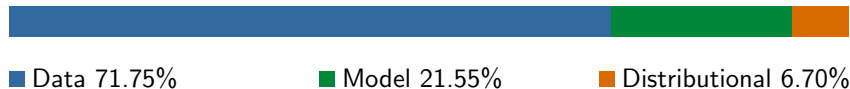
- Neural networks are widely used in vision, language, speech, etc.
- Despite strong performance, their predictions can be overconfident.
- Estimating uncertainty in a model's predictions can support safer decisions.
- Especially important when errors have serious consequences.
  - ▶ Autonomous driving, medical, financial, legal, etc.



- For a classification task, a neural network outputs probabilities for each class label.



- Predictive uncertainty* quantifies uncertainty about the class label of an input.
  - Does not by itself guide decisions about how to act.
  - Where does the predictive uncertainty come from?



- **Aleatoric (data) uncertainty** irreducible uncertainty from complexity in the data.
  - ▶ Defer the decision if high risk.
- **Epistemic (model) uncertainty** reducible uncertainty in the estimated network weights.
  - ▶ Collect more relevant training examples.
- **Distributional uncertainty** is uncertainty about the class probabilities at fixed network weights.
  - ▶ If inputs differ from training data, collect representative training examples.
- **Different uncertainty terms require different actions.**

- **Bayesian neural networks**

- ▶ Model uncertainty over network weights.
- ▶ Aleatoric and epistemic uncertainty.

- **Deterministic Dirichlet networks**

- ▶ Model a distribution over class probabilities
- ▶ Aleatoric and distributional uncertainty.

- Neither decomposes into all three uncertainty terms.

- Model that represents aleatoric, distributional, and epistemic uncertainty.
  - ▶ Combine **Bayesian posterior** distribution over network weights with a **Dirichlet distribution** over class probabilities.
- Understand how the resulting predictive uncertainty is divided between the three uncertainty terms.

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- $C$  classes, input  $x$ , label  $y \in \{1, \dots, C\}$ , training data  $D = \{(x_i, y_i)\}_{i=1}^N$ .
- The network with weights  $w$  outputs class-probability vector,

$$f(x; w) = (f_1(x; w), \dots, f_C(x; w)), \quad f_c(x; w) \geq 0, \quad \sum_{c=1}^C f_c(x; w) = 1.$$

- Categorical likelihood and weight prior define the posterior over weights.

$$p(D \mid w) = \prod_{i=1}^N \text{Cat}(y_i \mid f(x_i; w)) = \prod_{i=1}^N f_{y_i}(x_i; w),$$

$$\underbrace{p(w \mid D)}_{\text{posterior}} \propto \underbrace{p(D \mid w)}_{\text{likelihood}} \underbrace{p(w)}_{\text{prior}}.$$

- Random class-probability vector  $\pi = (\pi_1, \dots, \pi_C)$ ,  $\pi_c \geq 0 \ \forall c$ ,  $\sum_{c=1}^C \pi_c = 1$ .
- $\pi$  is Dirichlet distributed with concentration parameters  $\alpha = (\alpha_1, \dots, \alpha_C)$ ,  $\alpha_c > 0 \ \forall c$ ,  
$$p(\pi \mid \alpha) = \text{Dir}(\pi \mid \alpha).$$
- Mean vector and random class-probability variance,

$$m = \mathbb{E}[\pi] = \frac{\alpha}{\alpha_0}, \quad \underbrace{\alpha_0 = \sum_{c=1}^C \alpha_c}_{\text{total concentration parameter}}$$

$$\text{Var}(\pi_c) = \frac{m_c(1 - m_c)}{\alpha_0 + 1}.$$

- At a fixed mean, larger total concentration gives less spread.

- Dirichlet-categorical network outputs  $\alpha(x; w)$ .

$$p(\pi \mid x, w) = \text{Dir}(\pi \mid \alpha(x; w)),$$
$$p(y \mid \pi) = \text{Cat}(y \mid \pi) = \pi_y.$$

- Marginalise  $\pi$  to obtain label probabilities for input  $x$ .

$$p(y \mid x, w) = \int p(y \mid \pi) p(\pi \mid x, w) d\pi$$
$$= \mathbb{E}_{\text{Dir}(\pi \mid \alpha(x; w))}[\pi_y] = \frac{\alpha_y(x; w)}{\alpha_0(x; w)} = m_y(x; w).$$

- **Evidential deep learning (EDL)** parameterises the concentrations through evidence  $e_c(x; w) > 0$ , with  $\alpha_c(x; w) = e_c(x; w) + 1$ .
  - ▶ Use subjective logic to interpret evidence as support for classes.
  - ▶ Zero evidence corresponds to uniform Dirichlet.

- **Predictive:** Uncertainty about class label under model's predicted probabilities.
- **Aleatoric:** Label uncertainty that remains once class probabilities are known.
- **Distributional:** Reduction in label uncertainty obtained by knowing  $\pi$ , given weights  $w$ .
  - ▶ *Categorical BNN*: Point mass  $P(\pi = f(x; w) | x, w) = 1$ .
  - ▶ Zero distributional uncertainty.
- **Epistemic:** Reduction in label uncertainty obtained by knowing the weights  $w$ .
  - ▶ *Deterministic Dirichlet network*: Point mass  $P(w = \hat{w}) = 1$ .
  - ▶ Zero epistemic uncertainty.
- **Combining weight posterior with a Dirichlet distribution over class probabilities represents all three uncertainty terms.**

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- **One observed label** per input.

- **Network outputs**  $\alpha(x; w) = e(x; w) + 1$ ,

$$p(\pi \mid x, w) = \text{Dir}(\pi \mid \alpha(x; w)),$$

$$p(y \mid \pi) = \text{Cat}(y \mid \pi).$$

- Marginalising  $\pi$  gives observation model  $p(y \mid x, w) = m_y(x; w)$ .

- Assuming conditionally independent labels, **likelihood** is

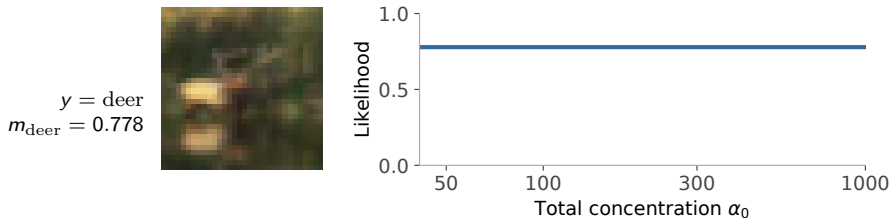
$$p(D \mid w) = \prod_{i=1}^N p(y_i \mid x_i, w) = \prod_{i=1}^N m_{y_i}(x_i; w).$$

# Single Labels Cannot Inform Total Concentration

- Single label likelihood **depends only on mean vector**,

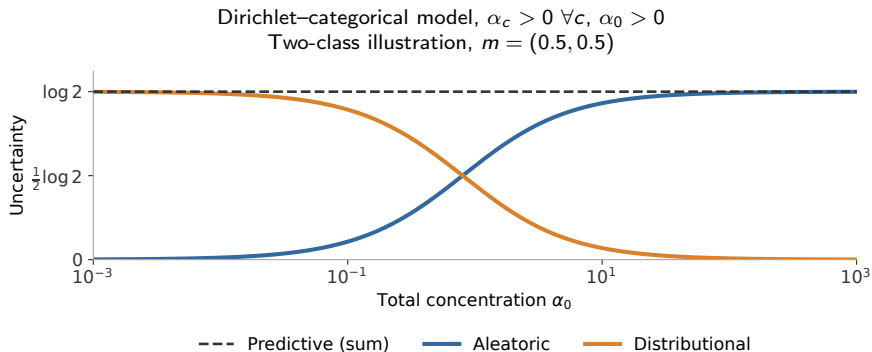
$$p(D \mid w) = \prod_{i=1}^N m_{y_i}(x_i; w) = \prod_{i=1}^N \frac{\alpha_{y_i}(x_i; w)}{\alpha_0(x_i; w)}.$$

- Likelihood **does not inform total concentration** because observed labels provide no evidence favouring one total concentration over another, at fixed mean vectors.



# Total Concentration Controls the Uncertainty Split

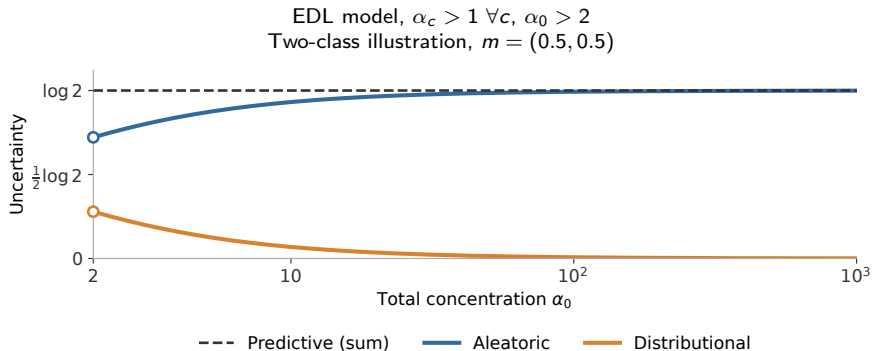
- At fixed mean vector, **total concentration** controls **aleatoric–distributional** allocation.



- Single labels **cannot inform** this uncertainty allocation.

# A Limitation of the Evidence Parameterisation

- EDL **restricts** the aleatoric–distributional split.
  - ▶ Fully distributional allocation cannot be approached.



- Would **less restrictive** parameterisation  $\alpha_c > 0 \forall c$  be preferable?

- At fixed mean vectors, single labels **do not inform total concentration**.
  - ▶ And therefore not the uncertainty allocation.
- State the **total concentration assumption** through prior,

$$\tilde{p}(w) \propto p(w) \prod_{i=1}^N \text{Gamma}(\alpha_0(x_i; w) \mid \kappa, \beta).$$

- ▶ Shape  $\kappa > 1$ , rate  $\beta > 0$ .
  - ▶ Prior mode  $(\kappa - 1)/\beta > C$  and standard deviation  $\sqrt{\kappa}/\beta$ .
- Prior  $\tilde{p}(w)$  and categorical likelihood give weight posterior,

$$p(w \mid D) \propto \tilde{p}(w) \prod_{i=1}^N m_{y_i}(x_i; w).$$

# Uncertainty Allocation Depends on Prior Assumption

- Vary **prior mode**, prior standard deviation 20% of mode.

| Prior mode           | Predictive entropy  | Aleatoric % | Distributional % | Epistemic % |
|----------------------|---------------------|-------------|------------------|-------------|
| <b>Fashion-MNIST</b> |                     |             |                  |             |
| 30                   | 1.336 (1.333–1.341) | 90.36       | 9.39             | 0.25        |
| 58                   | 0.934 (0.920–0.958) | 92.51       | 7.07             | 0.42        |
| 100                  | 0.974 (0.934–1.008) | 95.42       | 4.13             | 0.45        |
| 300                  | 1.947 (1.922–1.959) | 99.12       | 0.77             | 0.11        |
| <b>CIFAR-10</b>      |                     |             |                  |             |
| 30                   | 1.414 (1.399–1.436) | 89.24       | 8.98             | 1.78        |
| 100                  | 0.818 (0.813–0.822) | 91.11       | 4.79             | 4.10        |
| 300                  | 1.733 (1.728–1.739) | 98.07       | 0.87             | 1.06        |

- Higher prior mode  $\Rightarrow$  **larger aleatoric** and **smaller distributional** shares.
  - ▶ Uncertainty allocation depends on the **prior assumption**.
  - ▶ Prior assumption moves allocation as theory predicts.

- Compare predictive performance of Cat-eBNN against Cat-BNN.

| model                | accuracy            | NLL                 |
|----------------------|---------------------|---------------------|
| <b>Fashion-MNIST</b> |                     |                     |
| Cat-BNN              | 0.921 (0.917–0.924) | 0.221 (0.211–0.230) |
| Cat-eBNN             | 0.913 (0.908–0.916) | 0.431 (0.410–0.458) |
| <b>CIFAR-10</b>      |                     |                     |
| Cat-BNN              | 0.856 (0.850–0.859) | 0.426 (0.418–0.441) |
| Cat-eBNN             | 0.867 (0.866–0.868) | 0.470 (0.461–0.476) |

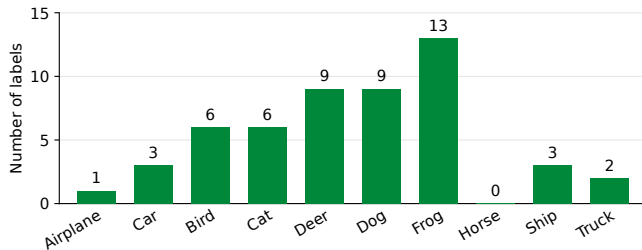
- Competitive, but **no consistent predictive improvement**.

- Cat-eBNN allow decomposition into all three uncertainty terms.
- At fixed means, single labels cannot inform total concentration.
  - ▶ Neither aleatoric-distributional allocation.
- Prior supplies the total concentration assumption that influences the allocation.
- Not clear practical value.
- **Can data inform uncertainty allocation?**

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- **Repeated labels** are multiple observed labels for same input.



- CIFAR-10H example with 52 human labels.
- A single selected label hides this **observed disagreement**.

- **Repeated labels** per input give counts  $n = (n_1, \dots, n_C)$ , with  $R = \sum_c n_c$ .

- **Network outputs**  $\alpha(x; w) = e(x; w) + 1$ .

$$p(\pi \mid x, w) = \text{Dir}(\pi \mid \alpha(x; w)),$$

$$p(n \mid \pi, R) = \text{Mult}(n \mid R, \pi).$$

- Marginalising  $\pi$  gives observation model  $p(n \mid x, w) = \text{DM}(n \mid R, \alpha(x; w))$ .

- For count data  $D_{\text{count}} = \{(x_i, n_i)\}_{i=1}^N$ , the **likelihood** is

$$p(D_{\text{count}} \mid w) = \prod_{i=1}^N p(n_i \mid x_i, w) = \prod_{i=1}^N \text{DM}(n_i \mid R_i, \alpha(x_i; w)).$$

# Repeated Labels Can Inform Total Concentration

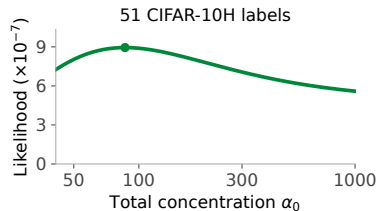
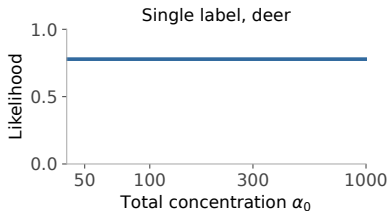
- Count likelihood **depends on total concentration**,

$$p(D_{\text{count}} | w) = \prod_{i=1}^N \left[ \frac{R_i!}{\prod_{c=1}^C n_{ic}!} \frac{\Gamma(\alpha_0(x_i; w))}{\Gamma(\alpha_0(x_i; w) + R_i)} \prod_{c=1}^C \frac{\Gamma(\alpha_0(x_i; w)m_c(x_i; w) + n_{ic})}{\Gamma(\alpha_0(x_i; w)m_c(x_i; w))} \right].$$

- Likelihood **can inform total concentration** because observed counts provide evidence favouring some total concentrations over others, at fixed mean vectors.
  - Count data can therefore inform aleatoric-distributional allocation.

Fixed mean  
 $m_{\text{deer}} = 0.778$

Observed counts  
39 deer, 5 ship  
7 other labels  
 $R = 51$



- **Multinomial BNN** (M-BNN) provides count prediction baseline,

$$p(D_{\text{count}} \mid w) = \prod_{i=1}^N \text{Mult}(n_i \mid R_i, f(x_i; w)).$$

- ▶ Given  $x$ ,  $w$ , class probability vector  $f(x; w)$  fixed.

| Model   | Predictive uncertainty | Aleatoric share (%) | Distributional share (%) | Epistemic share (%) |
|---------|------------------------|---------------------|--------------------------|---------------------|
| M-BNN   | 0.520 (0.503–0.532)    | 95.25               | –                        | 4.75                |
| DM-eBNN | 0.562 (0.554–0.573)    | 88.28               | 4.07                     | 7.65                |

- DM-eBNN can decompose into all **three uncertainty terms**.
- Counts **can inform** the aleatoric–distributional split.
- Does data driven decomposition support **better decisions**?

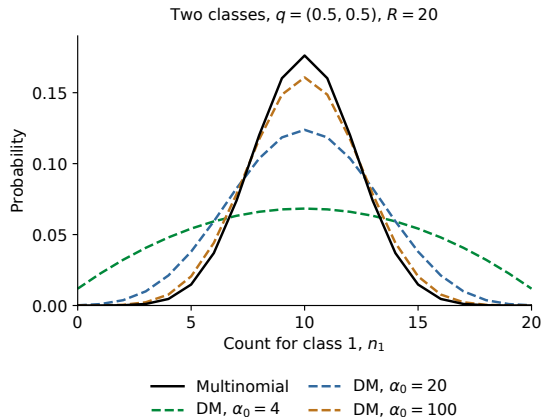
- Beyond uncertainty decomposition, evaluate **count prediction**.

| model            | majority label<br>accuracy | count NLL              |
|------------------|----------------------------|------------------------|
| <b>CIFAR-10H</b> |                            |                        |
| M-BNN            | 0.746 (0.722–0.776)        | 27.540 (23.630–31.685) |
| DM-eBNN          | 0.773 (0.759–0.781)        | 16.152 (15.664–16.860) |

- DM-eBNN **improves predictive fit**.
- Why?

# What Could Explain Predictive Fit Improvement?

- Compare count distributions of multinomial and Dirichlet multinomial.
- Set  $q$  = multinomial class-probability vector ( $f(x; w)$ ) = Dirichlet mean vector ( $m(x; w)$ ).
  - ▶ Count total  $R > 1$ .
- **Expected counts:**  $\mathbb{E}[n_c] = Rq_c$ .
- **Variance:**  $\text{Var}_{\text{DM}}[n_c] = \frac{\alpha_0 + R}{\alpha_0 + 1} \text{Var}_{\text{Mult}}[n_c]$ .
  - ▶ Total concentration controls additional count variance
- **Additional count variance** may explain the improved predictive fit.



- DM-eBNN represents all three uncertainty terms.
- At fixed means, count labels **can** inform total concentration.
  - ▶ Hence the aleatoric–distributional allocation.
- DM-eBNN improves held out count prediction in experiments.
- Better decisions from the uncertainty decomposition remain untested.

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- **EDL restricts concentrations to  $\alpha_c(x; w) > 1$ .**
  - ▶ Test direct positive concentrations,  $\alpha_c(x; w) > 0$ .
- **Practical value of uncertainty decomposition remains untested.**
  - ▶ Test whether the separate uncertainty terms improve decisions.
- **Training setup**
  - ▶ Multiple chains, larger sampling budgets, learning rate schedules and alternative samplers.
- **Count experiments cover only CIFAR-10H.**
  - ▶ Obtain and evaluate further datasets with repeated labels.

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- Cat-eBNN and DM-eBNN represent **all three uncertainty terms**.
- At fixed means, **total concentration** controls the aleatoric–distributional allocation.
  - ▶ Single labels cannot inform it. Cat-eBNN states a Gamma prior assumption.
  - ▶ Repeated labels can inform it.
- DM-eBNN improves **held out count prediction** over M-BNN in on CIFAR-10H.
- These findings motivate **collecting and retaining repeated labels** for uncertainty-aware classification.

Thank you for your attention :)

- For input  $x$  and weights  $w$ ,  $f(x; w)$  lists probabilities for  $C$  classes.
- $y$  is the label.  $\mathbb{I}[y = c]$  is 1 when  $y = c$  and 0 otherwise.

$$p(y \mid x, w) = \text{Cat}(y \mid f(x; w)) = \prod_{c=1}^C f_c(x; w)^{\mathbb{I}[y=c]} = f_y(x; w).$$

- $D$  contains  $N$  labelled pairs  $(x_i, y_i)$ .
- Conditional independence gives the likelihood as the product over observations.

$$p(D \mid w) = \prod_{i=1}^N p(y_i \mid x_i, w) = \prod_{i=1}^N f_{y_i}(x_i; w).$$

- For input  $x$ , class probabilities  $q = (q_1, \dots, q_C)$  satisfy  $q_c \geq 0$ ,  $\sum_c q_c = 1$ .
- Assume  $R$  independent categorical labels drawn with probabilities  $q$ .
  - ▶ Their counts are  $n = (n_1, \dots, n_C)$ , with  $n_c \in \{0, 1, \dots\}$  and  $\sum_c n_c = R$ .
- The count vector has a **multinomial distribution**.

$$p(n \mid R, q) = \text{Mult}(n \mid R, q) = \frac{R!}{\prod_{c=1}^C n_c!} \prod_{c=1}^C q_c^{n_c}.$$

- For  $R = 1$ , label  $y$  gives  $n_y = 1$  and all other counts zero.
  - ▶  $p(n \mid 1, q) = q_y = \text{Cat}(y \mid q)$ .

- For input  $x$ , random class probabilities  $\pi$  are **Dirichlet distributed**.

$$p(\pi \mid x, \alpha) = \text{Dir}(\pi \mid \alpha), \quad \alpha_c > 0, \quad \alpha_0 = \sum_{c=1}^C \alpha_c.$$

- Given  $\pi$ , the  $R$  labels are independent and their counts are **multinomial**,

$$p(n \mid R, \pi) = \text{Mult}(n \mid R, \pi).$$

- Marginalise**  $\pi$  to obtain count probability at input  $x$ .

$$\begin{aligned} p(n \mid x, R, \alpha) &= \text{DM}(n \mid R, \alpha) \\ &= \int_{\Delta^{C-1}} \text{Mult}(n \mid R, \pi) \text{Dir}(\pi \mid \alpha) d\pi \\ &= \frac{R!}{\prod_{c=1}^C n_c!} \frac{\Gamma(\alpha_0)}{\Gamma(\alpha_0 + R)} \prod_{c=1}^C \frac{\Gamma(\alpha_c + n_c)}{\Gamma(\alpha_c)}. \end{aligned}$$

- A (0.5, 0.5) bird–frog class probability prediction can reflect
  - ▶ **Little evidence** for either class.
  - ▶ **Evidence** that supports both classes equally.
- For  $C$  classes, subjective logic represents the model's belief about the class label using
  - ▶  $b_c$ , belief assigned to class  $c$ .
  - ▶  $u$ , belief not assigned to any individual class.

$$b_c \geq 0, \quad u \geq 0, \quad \sum_{c=1}^C b_c + u = 1.$$

- ▶  $a_c$ , base probability is fraction of unassigned belief allocated to class  $c$

$$a_c \geq 0, \quad \sum_{c=1}^C a_c = 1.$$

- Probability of class  $c$  is a function of its **assigned** and **unassigned** belief,

$$p(y = c) = \underbrace{b_c}_{\text{assigned belief}} + \underbrace{a_c u}_{\text{share of unassigned belief}}$$

- Consider  $C = 2$  classes, bird ( $c = 1$ ) and frog ( $c = 2$ ).
  - Base probabilities**  $a_c$  distribute unassigned belief. Choose  $a_1 = a_2 = 0.5$ .
- No assigned belief**
  - Class belief**  $b_c$  assigned to class  $c$ . No belief assigned  $b_1 = b_2 = 0$ .
  - Unassigned belief**  $u = 1$  (since  $\sum_c b_c + u = 1$ ).

$$p(y = c) = b_c + a_c u = 0 + 0.5 \times 1 = 0.5, \quad c \in \{1, 2\}.$$

- $(0.5, 0.5)$  predicted class probability reflects **no evidence supporting belief for either class**.
  - Belief assigned**
    - Suppose **evidence supports assigning belief** to each class.  $b_1 = b_2 = 0.5$  and  $u = 0$ .
- $$p(y = c) = b_c + a_c u = 0.5 + 0.5 \times 0 = 0.5, \quad c \in \{1, 2\}.$$
- $(0.5, 0.5)$  predicted class probability reflects **evidence supporting belief for equal classes**.

- The network outputs evidence  $e_c(x; w) \geq 0$ , **learned support for class  $c$** .
- EDL chooses a **uniform Dirichlet distribution at zero evidence**.
  - ▶ Baseline concentration of one for every class.
- Learned evidence adds to this baseline,

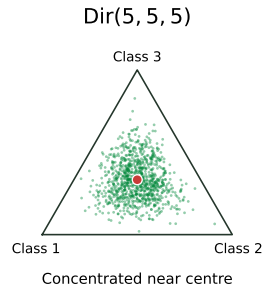
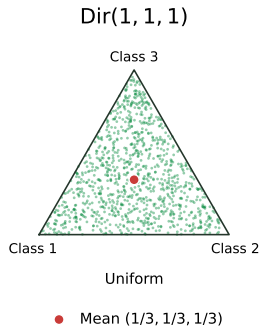
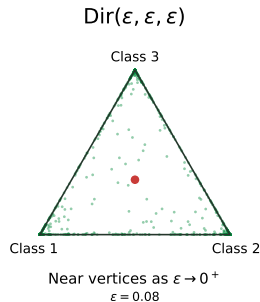
$$\alpha_c(x; w) = \underbrace{1}_{\text{baseline}} + \underbrace{e_c(x; w)}_{\text{learned class evidence}}.$$

- Concentration parameters specify a distribution over random class probability vectors  $\pi$ ,

$$p(\pi \mid x, w) = \text{Dir}(\pi \mid \alpha(x; w)).$$

# The Dirichlet Baseline at Zero Evidence

- Each dot is a sampled class-probability vector.
  - Vertices assign probability one to a class.
  - The centre assigns  $1/3$  probability to each class.



- EDL chooses **uniform density at zero evidence**

- Following Sensoy et al. (2018) normalised evidence gives **assigned class belief**, normalised baseline gives **unassigned belief**.
  - ▶ Normalise by  $\alpha_0(x; w) = \sum_c \alpha_c(x; w)$  gives belief masses,

$$b_c(x; w) = \frac{e_c(x; w)}{\alpha_0(x; w)},$$

$$u(x; w) = 1 - \sum_{c=1}^C b_c(x; w) = \frac{C}{\alpha_0(x; w)}.$$

- ▶  $a_c = 1/C$  base probabilities  $\forall c$  since each class has one baseline unit.
- Gives **subjective logic interpretation** of Dirichlet–categorical observation model,

$$\underbrace{p(y = c \mid x, w)}_{\text{observation model}} = \underbrace{b_c(x; w) + a_c u(x; w)}_{\text{subjective logic}} = \underbrace{\frac{\alpha_c(x; w)}{\alpha_0(x; w)}}_{\text{Dirichlet mean}}.$$

- **Entropy** measures label uncertainty,

$$\mathcal{H}[p(y)] = - \sum_{c=1}^C p(y=c) \log p(y=c).$$

- **Expected conditional entropy** averages what remains when the random variable  $\xi$  is known,

$$\mathcal{H}[y \mid \xi] = \mathbb{E}_{p(\xi)}[\mathcal{H}[p(y \mid \xi)]] .$$

- **Mutual information** is the average reduction from knowing  $\xi$ ,

$$\begin{aligned} \mathcal{I}(y; \xi) &= \mathcal{H}[p(y)] - \mathcal{H}[y \mid \xi], \\ \underbrace{\mathcal{H}[p(y)]}_{\text{total}} &= \underbrace{\mathcal{H}[y \mid \xi]}_{\text{average remaining}} + \underbrace{\mathcal{I}(y; \xi)}_{\text{average reduction}}. \end{aligned}$$

- Apply the decomposition to weights  $w$ , then to class probabilities  $\pi$  given  $w$ .
- Given input  $x$  and training data  $D$ ,

$$\underbrace{\mathcal{H}[p(y \mid x, D)]}_{\text{predictive}} = \underbrace{\mathbb{E}_{p(w|D)} [\mathbb{E}_{p(\pi|x,w)} [\mathcal{H}[\text{Cat}(y \mid \pi)]]]}_{\text{aleatoric}} \\ + \underbrace{\mathbb{E}_{p(w|D)} [\mathcal{I}(y; \pi \mid w)]}_{\text{distributional}} \\ + \underbrace{\mathcal{I}(y; w)}_{\text{epistemic}}.$$

- **Categorical BNN** Given  $x, w$ ,  $p(\pi = f(x; w) | x, w) = 1$ .

Distributional term is zero.

$$\underbrace{\mathcal{H}[p(y | x, D)]}_{\text{predictive}} = \underbrace{\mathbb{E}_{p(w|D)}[\mathcal{H}[\text{Cat}(y | f(x; w))]]}_{\text{aleatoric}} + \underbrace{\mathcal{I}(y; w)}_{\text{epistemic}}.$$

- **Deterministic Dirichlet network** The weights are fixed at  $\hat{w}$ .

Epistemic term is zero.

$$\underbrace{\mathcal{H}[p(y | x, \hat{w})]}_{\text{predictive}} = \underbrace{\mathbb{E}_{\text{Dir}(\pi | \alpha(x; \hat{w}))}[\mathcal{H}[\text{Cat}(y | \pi)]]}_{\text{aleatoric}} + \underbrace{\mathcal{I}(y; \pi | \hat{w})}_{\text{distributional}}.$$

- Combining a **weight posterior** with a **Dirichlet distribution** for class probabilities allows all three terms.

- $p(\pi \mid \alpha) = \text{Dir}(\pi \mid \alpha)$ ,  $p(y \mid \pi) = \text{Cat}(y \mid \pi)$ ,  $\alpha_c > 0$ .

- Marginalising  $\pi$  gives  $p(y \mid \alpha) = m_y = \text{Cat}(y \mid m)$

►  $m = \alpha / \alpha_0$ ,  $\alpha_0 = \sum_c \alpha_c$

- Predictive uncertainty decomposes as

$$\underbrace{\mathcal{H}[\text{Cat}(y \mid m)]}_{\text{predictive}} = \underbrace{\mathbb{E}_{\text{Dir}(\pi \mid \alpha)}[\mathcal{H}[\text{Cat}(y \mid \pi)]]}_{\text{aleatoric}} + \underbrace{\mathcal{I}(y; \pi)}_{\text{distributional}}.$$

- Closed form aleatoric term with  $\alpha = \alpha_0 m$ ,

$$A_m(\alpha_0) := \mathbb{E}_{\text{Dir}(\pi \mid \alpha_0 m)}[\mathcal{H}[\text{Cat}(y \mid \pi)]] = \sum_{c=1}^C m_c [\psi(\alpha_0 + 1) - \psi(\alpha_0 m_c + 1)],$$

- $\psi(t) = \frac{d}{dt} \log \Gamma(t)$  is the digamma function.

## Proposition 2.1. Aleatoric Monotonicity

- Fix Dirichlet mean vector  $m$ ,  $0 < m_c < 1 \forall c$ .
- **Proposition:**  $A_m(\alpha_0)$  is strictly increasing for  $\alpha_0 > 0$ .
- **Proof** summary: Differentiate at fixed  $m$ .

$$A'_m(\alpha_0) = \sum_{c=1}^C m_c [\psi'(\alpha_0 + 1) - m_c \psi'(\alpha_0 m_c + 1)].$$

- ▶ The function  $g(t) = t\psi'(t+1)$  is strictly increasing for  $t > 0$  (Li et al. 2008, Thm. 1).
- ▶ Since  $0 < \alpha_0 m_c < \alpha_0$ ,  $g(\alpha_0 m_c) < g(\alpha_0)$ , so

$$\psi'(\alpha_0 + 1) - m_c \psi'(\alpha_0 m_c + 1) > 0.$$

$$A'_m(\alpha_0) > 0.$$

- Fix Dirichlet mean vector  $m$ ,  $0 < m_c < 1 \forall c$ .

$$A_m(\alpha_0) = \sum_{c=1}^C m_c [\psi(\alpha_0 + 1) - \psi(\alpha_0 m_c + 1)].$$

- **Proposition:**

$$\lim_{\alpha_0 \rightarrow 0^+} A_m(\alpha_0) = 0, \quad \lim_{\alpha_0 \rightarrow \infty} A_m(\alpha_0) = \mathcal{H}[\text{Cat}(y \mid m)].$$

- **Proof** summary:

- ▶ As  $\alpha_0 \rightarrow 0^+$ , continuity gives  $\psi(\alpha_0 + 1) - \psi(\alpha_0 m_c + 1) \rightarrow 0$ .
- ▶ As  $z \rightarrow \infty$ ,  $\psi(z) = \log z + \mathcal{O}(z^{-1})$  (e.g. Abramowitz and Stegun 1972, Eq. 6.4.1) gives  $\psi(\alpha_0 + 1) - \psi(\alpha_0 m_c + 1) \rightarrow -\log m_c$ .

## Corollary 2.3. Distributional Uncertainty

- Fix Dirichlet mean vector  $m$ ,  $0 < m_c < 1 \forall c$ .

$$\mathcal{I}(y; \pi) = \mathcal{H}[\text{Cat}(y \mid m)] - A_m(\alpha_0).$$

- **Corollary:**  $\mathcal{I}(y; \pi)$  is strictly decreasing for  $\alpha_0 > 0$ , and

$$\lim_{\alpha_0 \rightarrow 0^+} \mathcal{I}(y; \pi) = \mathcal{H}[\text{Cat}(y \mid m)], \quad \lim_{\alpha_0 \rightarrow \infty} \mathcal{I}(y; \pi) = 0.$$

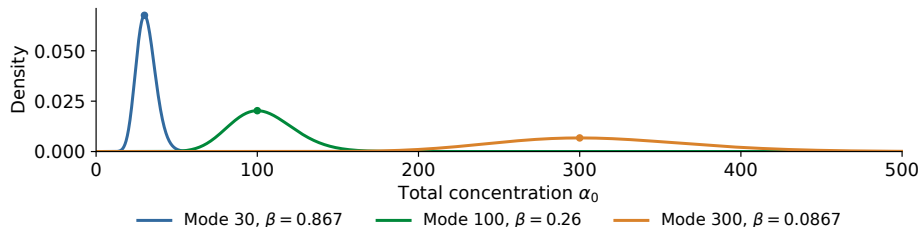
- **Proof** summary:

- ▶ At fixed  $m$ ,  $\mathcal{H}[\text{Cat}(y \mid m)]$  is constant. Proposition 2.1 gives  $A_m(\alpha_0)$  strictly increasing, so  $\mathcal{I}(y; \pi)$  strictly decreasing.
- ▶ Proposition 2.2 gives  $A_m(\alpha_0) \rightarrow 0$  as  $\alpha_0 \rightarrow 0^+$  and  $A_m(\alpha_0) \rightarrow \mathcal{H}[\text{Cat}(y \mid m)]$  as  $\alpha_0 \rightarrow \infty$ . Subtracting these limits from  $\mathcal{H}[\text{Cat}(y \mid m)]$  gives limits for  $\mathcal{I}(y; \pi)$ .

$$\text{Gamma}(t \mid \kappa, \beta) = \frac{\beta^\kappa}{\Gamma(\kappa)} t^{\kappa-1} e^{-\beta t}, \quad t > 0, \quad \kappa > 0, \quad \beta > 0,$$

$$\text{Mode} = \frac{\kappa - 1}{\beta} \quad (\kappa > 1), \quad \text{SD} = \frac{\sqrt{\kappa}}{\beta}.$$

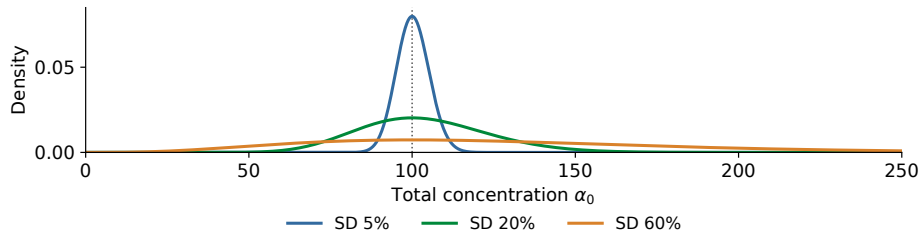
- **Fix shape**  $\kappa = 27$ , so SD stays approximately 20% of the prior mode.
- **Vary rate**  $\beta = 26/\bar{\alpha}_0$  to set prior modes  $\bar{\alpha}_0 \in \{30, 100, 300\}$ .



$$\text{Gamma}(t \mid \kappa, \beta) = \frac{\beta^\kappa}{\Gamma(\kappa)} t^{\kappa-1} e^{-\beta t}, \quad t > 0, \quad \kappa > 0, \quad \beta > 0,$$

$$\text{Mode} = \frac{\kappa - 1}{\beta} \quad (\kappa > 1), \quad \text{SD} = \frac{\sqrt{\kappa}}{\beta}.$$

- **Fix prior mode**  $\bar{\alpha}_0 = 100$ , so  $\beta = (\kappa - 1)/100$ .
- **Vary shape and rate** to set SD to 5%, 20% and 60% of the prior mode.



- For  $C = 10$ , the EDL parameterisation gives

$$\alpha_0(x; w) = 10 + \sum_{c=1}^{10} e_c(x; w) > 10.$$

- Approaching the lower bound forces **uniform predicted class probabilities**.

$$\alpha_0(x; w) \rightarrow 10^+ \implies \alpha_c(x; w) \rightarrow 1 \implies m_c(x; w) \rightarrow \frac{1}{10} \quad \forall c.$$

- **Fashion-MNIST**, ten classes.
  - ▶ 54000 training, 6000 validation, 10000 test images.
- **CIFAR-10**, ten classes.
  - ▶ 45000 training, 5000 validation, 10000 test images.
- **CIFAR-10H**, repeated labels for the CIFAR-10 test images.
  - ▶ Stratified split of 8000 training, 1000 validation, 1000 test images.
  - ▶ 47–63 labels per image, mean 51.1. Aggregate by class.

- ResNet-20, filter response normalisation.
- Training augmentation: random crops, horizontal flips and per-channel normalisation.
- Three seeds.
- One SGLD chain per seed.

|                                  | Fashion-MNIST | CIFAR-10 | CIFAR-10H                      |
|----------------------------------|---------------|----------|--------------------------------|
| Cold SGLD epochs                 | 150           | 600      | 3000                           |
| Deterministic pretraining epochs | 50            | 300      | 400                            |
| Warm SGLD epochs                 | 100           | 300      | 1000                           |
| Constant SGLD learning rate      | 0.01          | 0.01     | M-BNN 0.0002<br>DM-eBNN 0.0012 |
| Retained weight samples          | 10            | 10       | 40                             |
| Sample spacing in epochs         | 5             | 20       | 20                             |

- $M$  evaluation inputs.  $S$  retained weight samples  $w^{(s)}$ .

- **Probability of the observed label**

$$p(y_i | x_i, w^{(s)}) = \begin{cases} f_{y_i}(x_i; w^{(s)}), & \text{Softmax, Cat-BNN,} \\ m_{y_i}(x_i; w^{(s)}), & \text{EDL, Cat-eBNN.} \end{cases}$$

- **Negative log likelihood**, averaging probabilities before taking logarithms.

$$\text{NLL} = -\frac{1}{M} \sum_{i=1}^M \log \left[ \frac{1}{S} \sum_{s=1}^S p(y_i | x_i, w^{(s)}) \right].$$

- Deterministic models use  $S = 1$  and  $w^{(1)} = \hat{w}$ .

- $M$  evaluation inputs.  $S$  weight samples. Count total  $R_i = \sum_c n_{ic}$ .

- **Probability of the observed count vector**

$$p(n_i \mid x_i, R_i, w^{(s)}) = \begin{cases} \text{Mult}(n_i \mid R_i, f(x_i; w^{(s)})), & \text{M-BNN}, \\ \text{DM}(n_i \mid R_i, \alpha(x_i; w^{(s)})), & \text{DM-eBNN}, \\ \text{Mult}(n_i \mid R_i, m(x_i; w^{(s)})), & \text{mean matched control.} \end{cases}$$

- **Count NLL**, using the corresponding probability above.

$$\text{NLL}_{\text{count}} = -\frac{1}{M} \sum_{i=1}^M \log \left[ \frac{1}{S} \sum_{s=1}^S p(n_i \mid x_i, R_i, w^{(s)}) \right].$$

- The control retains DM-eBNN mean vectors and weight samples. No retraining.

- **Cold initialisation** uses random weights.
- **Warm initialisation** loads the best validation checkpoint,
  - ▶ Cat-BNN from Softmax,
  - ▶ Cat-eBNN from EDL,
  - ▶ Count models use matching count likelihoods.
- **Shifted initialisation** adjusts the warm EDL output biases.
  - ▶ Match the initial median total concentration of training subset to the prior mode.
  - ▶ Predicted classes are preserved.
  - ▶ Class probabilities can change.

- Add  $b$  to every output bias of  $w_{\text{warm}}$ , giving weights  $w_b$ .

$$\alpha_0(x; w_b) = \sum_{c=1}^C \text{softplus}(z_c(x; w_{\text{warm}}) + b) + C.$$

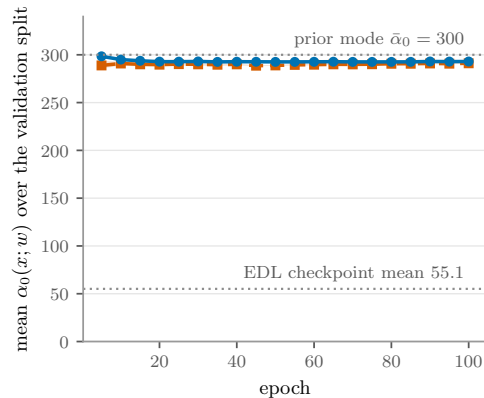
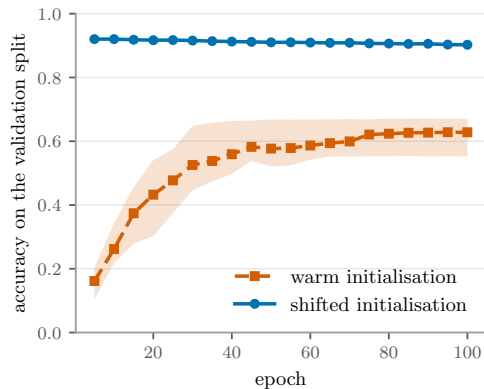
- Match median over subset of training inputs  $P$  to the prior mode  $\bar{\alpha}_0$ .

$$\text{median}_{x \in P} \left[ \sum_{c=1}^C \text{softplus}(z_c(x; w_{\text{warm}}) + b) + C \right] = \bar{\alpha}_0.$$

- Solve for  $b$  with **Bisection**:  $b_{\text{low}}$ ,  $b_{\text{high}}$  and tests  $b_{\text{mid}} = (b_{\text{low}} + b_{\text{high}})/2$ .
  - ▶ Median concentration below  $\bar{\alpha}_0 \Rightarrow$  set  $b_{\text{low}} = b_{\text{mid}}$ .
  - ▶ Median concentration above  $\bar{\alpha}_0 \Rightarrow$  set  $b_{\text{high}} = b_{\text{mid}}$ .
  - ▶ Repeat until close to  $\bar{\alpha}_0$ .
  - ▶ Apply the final shift  $b$  once at initialisation before starting SGLD.

# Experimental Setup: Shifted vs Warm Initialisation Diagnostics

- Initialisation affects predictive performance within budget.



| BMA accuracy by initialisation method, validation split |                  |                     |                            |                            |
|---------------------------------------------------------|------------------|---------------------|----------------------------|----------------------------|
| model                                                   | Gamma prior mode | cold                | warm                       | shifted                    |
| <b>Fashion-MNIST</b>                                    |                  |                     |                            |                            |
| Cat-BNN                                                 | –                | 0.871 (0.868–0.874) | <b>0.918 (0.913–0.926)</b> | –                          |
| Cat-eBNN                                                | 30               | 0.689 (0.654–0.707) | <b>0.920 (0.915–0.925)</b> | 0.855 (0.853–0.856)        |
| Cat-eBNN                                                | 58               | 0.477 (0.472–0.481) | 0.899 (0.856–0.924)        | <b>0.920 (0.916–0.924)</b> |
| Cat-eBNN                                                | 100              | 0.230 (0.200–0.277) | 0.875 (0.854–0.917)        | <b>0.916 (0.912–0.920)</b> |
| Cat-eBNN                                                | 300              | 0.152 (0.116–0.180) | 0.616 (0.553–0.671)        | <b>0.908 (0.905–0.910)</b> |
| <b>CIFAR-10</b>                                         |                  |                     |                            |                            |
| Cat-BNN                                                 | –                | 0.668 (0.658–0.680) | <b>0.856 (0.850–0.860)</b> | –                          |
| Cat-eBNN                                                | 30               | omitted             | <b>0.839 (0.790–0.863)</b> | 0.794 (0.767–0.831)        |
| Cat-eBNN                                                | 100              | 0.231 (0.225–0.239) | <b>0.866 (0.865–0.868)</b> | <b>0.866 (0.864–0.870)</b> |
| Cat-eBNN                                                | 300              | omitted             | 0.831 (0.779–0.857)        | <b>0.856 (0.854–0.857)</b> |

- Prior SD is 20% of the mode.
- Bold mark selected initialisations.
- **Warm and shifted initialisation** outperform cold initialisation within tested budget.

| Single-label predictive performance, test split |                     |                     |
|-------------------------------------------------|---------------------|---------------------|
| model                                           | accuracy            | NLL                 |
| <b>Fashion-MNIST</b>                            |                     |                     |
| Softmax                                         | 0.928 (0.923–0.933) | 0.202 (0.190–0.214) |
| EDL                                             | 0.924 (0.919–0.926) | 0.496 (0.477–0.520) |
| Cat-BNN                                         | 0.921 (0.917–0.924) | 0.221 (0.211–0.230) |
| Cat-eBNN                                        | 0.913 (0.908–0.916) | 0.431 (0.410–0.458) |
| <b>CIFAR-10</b>                                 |                     |                     |
| Softmax                                         | 0.879 (0.875–0.882) | 0.554 (0.512–0.595) |
| EDL                                             | 0.885 (0.884–0.887) | 0.530 (0.517–0.544) |
| Cat-BNN                                         | 0.856 (0.850–0.859) | 0.426 (0.418–0.441) |
| Cat-eBNN                                        | 0.867 (0.866–0.868) | 0.470 (0.461–0.476) |

- Cat-eBNN uses shifted initialisation, prior mode 100 and prior SD 20% of the mode.
- Cat-BNN uses warm initialisation.
- Deterministic models use best validation checkpoint.
- Cat-eBNN achieves **competitive accuracy**.
- **Lower NLL than EDL, but higher NLL than Cat-BNN.**

Gamma prior mode experiment at standard deviation 20%, test split

| Gamma prior mode     | realised total concentration | predictive uncertainty | aleatoric uncertainty % | distributional uncertainty % | epistemic uncertainty % | NLL                 |
|----------------------|------------------------------|------------------------|-------------------------|------------------------------|-------------------------|---------------------|
| <b>Fashion-MNIST</b> |                              |                        |                         |                              |                         |                     |
| 30                   | 30.5 (30.5–30.6)             | 1.336 (1.333–1.341)    | 90.36                   | 9.39                         | 0.25                    | 0.535 (0.529–0.545) |
| 58                   | 58.4 (58.4–58.5)             | 0.934 (0.920–0.958)    | 92.51                   | 7.07                         | 0.42                    | 0.382 (0.373–0.399) |
| 100                  | 99.0 (98.8–99.3)             | 0.974 (0.934–1.008)    | 95.42                   | 4.13                         | 0.45                    | 0.431 (0.410–0.458) |
| 300                  | 292.7 (292.3–292.9)          | 1.947 (1.922–1.959)    | 99.12                   | 0.77                         | 0.11                    | 1.143 (1.111–1.168) |
| <b>CIFAR-10</b>      |                              |                        |                         |                              |                         |                     |
| 30                   | 30.4 (30.2–30.5)             | 1.414 (1.399–1.436)    | 89.24                   | 8.98                         | 1.78                    | 0.763 (0.682–0.923) |
| 100                  | 99.7 (99.5–99.8)             | 0.818 (0.813–0.822)    | 91.11                   | 4.79                         | 4.10                    | 0.470 (0.461–0.476) |
| 300                  | 291.4 (291.2–291.6)          | 1.733 (1.728–1.739)    | 98.07                   | 0.87                         | 1.06                    | 0.916 (0.907–0.921) |

- Prior SD is fixed at 20% of the mode.
- Mode 30 uses warm initialisation. Higher modes use shifted initialisation.
- Higher prior modes increase realised total concentration
  - **Shifts allocation** towards aleatoric uncertainty.

Gamma prior standard deviation experiment at mode 100, test split

| Gamma prior standard deviation | realised total concentration | predictive uncertainty | aleatoric uncertainty % | distributional uncertainty % | epistemic uncertainty % | NLL                 |
|--------------------------------|------------------------------|------------------------|-------------------------|------------------------------|-------------------------|---------------------|
| <b>Fashion-MNIST</b>           |                              |                        |                         |                              |                         |                     |
| 1000%                          | 67.6 (66.8–68.8)             | 0.854 (0.836–0.874)    | 92.40                   | 7.11                         | 0.49                    | 0.359 (0.348–0.374) |
| 60%                            | 90.4 (89.6–91.3)             | 0.903 (0.879–0.935)    | 94.74                   | 4.85                         | 0.41                    | 0.393 (0.378–0.418) |
| 20%                            | 99.0 (98.8–99.3)             | 0.974 (0.934–1.008)    | 95.42                   | 4.13                         | 0.45                    | 0.431 (0.410–0.458) |
| 5%                             | 100.1 (100.1–100.2)          | 1.029 (0.988–1.061)    | 95.68                   | 3.90                         | 0.42                    | 0.454 (0.433–0.482) |
| <b>CIFAR-10</b>                |                              |                        |                         |                              |                         |                     |
| 1000%                          | 83.7 (82.2–85.0)             | 0.815 (0.812–0.818)    | 89.39                   | 6.03                         | 4.58                    | 0.473 (0.465–0.477) |
| 60%                            | 97.2 (97.1–97.3)             | 0.800 (0.798–0.803)    | 90.56                   | 5.03                         | 4.41                    | 0.466 (0.458–0.472) |
| 20%                            | 99.7 (99.5–99.8)             | 0.818 (0.813–0.822)    | 91.11                   | 4.79                         | 4.10                    | 0.470 (0.461–0.476) |
| 5%                             | 100.1 (100.0–100.1)          | 0.919 (0.892–0.934)    | 92.57                   | 4.32                         | 3.11                    | 0.508 (0.499–0.516) |

- Prior mode is fixed at 100. Shifted initialisation.
- Prior SD is varied as a percentage of the mode.
- Narrower priors push total concentration closer to the mode
  - ▶ Resulting in higher total concentration.
  - ▶ **Shifts allocation** towards aleatoric uncertainty.

| Predictive uncertainty and its decomposition, test split |                        |                       |                            |                       |
|----------------------------------------------------------|------------------------|-----------------------|----------------------------|-----------------------|
| model                                                    | predictive uncertainty | aleatoric uncertainty | distributional uncertainty | epistemic uncertainty |
| <b>Fashion-MNIST</b>                                     |                        |                       |                            |                       |
| Softmax                                                  | 0.183                  | 0.183 (100.00%)       | –                          | –                     |
| EDL                                                      | 1.038                  | 0.926 (89.21%)        | 0.112 (10.79%)             | –                     |
| Cat-BNN                                                  | 0.233                  | 0.225 (96.55%)        | –                          | 0.008 (3.45%)         |
| Cat-eBNN                                                 | 0.974                  | 0.929 (95.42%)        | 0.040 (4.13%)              | 0.004 (0.45%)         |
| <b>CIFAR-10</b>                                          |                        |                       |                            |                       |
| Softmax                                                  | 0.110                  | 0.110 (100.00%)       | –                          | –                     |
| EDL                                                      | 0.815                  | 0.732 (89.77%)        | 0.083 (10.23%)             | –                     |
| Cat-BNN                                                  | 0.390                  | 0.318 (81.38%)        | –                          | 0.073 (18.62%)        |
| Cat-eBNN                                                 | 0.818                  | 0.746 (91.11%)        | 0.039 (4.79%)              | 0.034 (4.10%)         |

- Averages over test inputs and three seeds. Parentheses give uncertainty shares.
- Cat-eBNN uses shifted initialisation, prior mode 100 and prior SD 20% of the mode.
- Cat-BNN uses warm initialisation.
- Cat-eBNN represents **all three uncertainty terms**,
  - ▶ Its aleatoric–distributional allocation **depends on the prior assumption**.

| model               | Gamma prior mode | Count prediction, test split |                         |                        |                          |
|---------------------|------------------|------------------------------|-------------------------|------------------------|--------------------------|
|                     |                  | realised total concentration | majority label accuracy | count NLL              | mean matched control NLL |
| Cold initialisation |                  |                              |                         |                        |                          |
| M-BNN               | –                | –                            | 0.724 (0.712–0.744)     | 19.161 (18.012–19.754) | –                        |
| DM-eBNN             | 50               | 50.0 (49.6–50.6)             | 0.712 (0.697–0.731)     | 12.718 (12.486–13.007) | 23.719 (23.195–24.467)   |
| DM-eBNN             | 200              | 178.6 (176.4–180.3)          | 0.702 (0.693–0.715)     | 13.199 (11.872–14.051) | 20.291 (17.602–22.035)   |
| Warm initialisation |                  |                              |                         |                        |                          |
| M-BNN               | –                | –                            | 0.746 (0.722–0.776)     | 27.540 (23.630–31.685) | –                        |
| DM-eBNN             | 50               | 51.0 (50.8–51.1)             | 0.753 (0.720–0.777)     | 14.392 (13.724–15.529) | 28.802 (26.771–32.415)   |
| DM-eBNN             | 200              | 182.7 (181.5–184.8)          | 0.773 (0.759–0.781)     | 16.152 (15.664–16.860) | 26.207 (25.281–27.394)   |

- DM-eBNN prior SD is 20% of mode.
- Cold SGLD 3000 epochs, warm SGLD 1000 epochs after 400 pretraining.
- DM-eBNN **lower count NLL** than M-BNN and mean matched controls.

| Gamma prior sensitivity, test split |                                                  |                                      |                                           |                              |                         |                        |
|-------------------------------------|--------------------------------------------------|--------------------------------------|-------------------------------------------|------------------------------|-------------------------|------------------------|
| initialisation                      | pretraining Gamma prior mode, standard deviation | initial realised total concentration | SGLD Gamma prior mode, standard deviation | realised total concentration | majority label accuracy | count NLL              |
| cold                                | none                                             | –                                    | none                                      | 10.0 (10.0–10.0)             | 0.099 (0.098–0.100)     | 23.433 (23.433–23.433) |
| warm                                | 50, 20%                                          | 52.1 (51.6–52.6)                     | 50, 20%                                   | 51.0 (50.8–51.1)             | 0.753 (0.720–0.777)     | 14.392 (13.724–15.529) |
| warm                                | 50, 20%                                          | 52.1 (51.6–52.6)                     | 50, 100%                                  | 69.8 (69.2–70.3)             | 0.754 (0.723–0.776)     | 13.626 (12.863–14.940) |
| warm                                | 50, 20%                                          | 52.1 (51.6–52.6)                     | none                                      | 79.3 (78.2–80.1)             | 0.752 (0.724–0.776)     | 13.657 (12.863–15.032) |
| warm                                | 200, 20%                                         | 186.7 (184.4–189.9)                  | 200, 20%                                  | 182.7 (181.5–184.8)          | 0.773 (0.759–0.781)     | 16.152 (15.664–16.860) |
| warm                                | 200, 20%                                         | 186.7 (184.4–189.9)                  | 200, 100%                                 | 162.5 (160.6–165.5)          | 0.771 (0.758–0.781)     | 14.439 (13.983–15.153) |
| warm                                | 200, 20%                                         | 186.7 (184.4–189.9)                  | none                                      | 155.9 (154.1–159.0)          | 0.772 (0.760–0.784)     | 13.941 (13.557–14.574) |

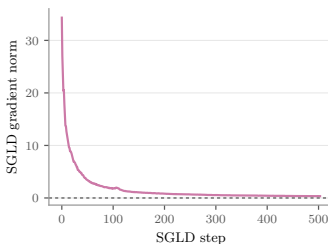
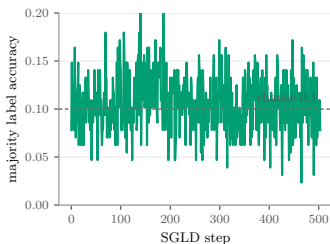
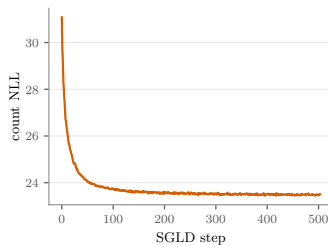
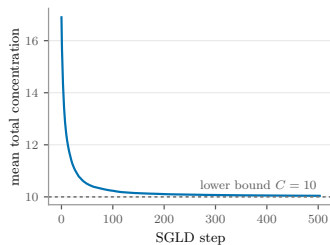
- Warm pretraining uses prior modes 50 or 200, with SD 20% of the mode.
- During SGLD, prior SD is 20% or 100% of the mode, or no Gamma prior.
- Wider or absent priors result in more **data informed total concentration** and improve count NLL.
- Cold training without a Gamma prior collapses.

| model   | Predictive uncertainty of the class label, test split |                        |                         |                              |                         |
|---------|-------------------------------------------------------|------------------------|-------------------------|------------------------------|-------------------------|
|         | Gamma prior mode                                      | predictive uncertainty | aleatoric uncertainty % | distributional uncertainty % | epistemic uncertainty % |
| M-BNN   | –                                                     | 0.520 (0.503–0.532)    | 95.25                   | –                            | 4.75                    |
| DM-eBNN | 50                                                    | 1.040 (1.036–1.047)    | 87.60                   | 7.71                         | 4.69                    |
| DM-eBNN | 200                                                   | 0.562 (0.554–0.573)    | 88.28                   | 4.07                         | 7.65                    |

- Averages over test inputs and three seeds. Parentheses give uncertainty shares.
- Warm initialisation.
- DM-eBNN prior modes 50 and 200, with SD 20% of the mode.
- DM-eBNN represents **all three uncertainty** terms,
  - ▶ Counts are **able to inform** the aleatoric–distributional split.

# Training without the Gamma Prior

- Count data can inform total concentration. Can we remove Gamma prior?



- Early updates lower count NLL by reducing total concentration.
  - As  $\alpha_0(x; w) \rightarrow 10$ , mean vectors become uniform.
  - Accuracy stays near chance.
- Evidence approaches zero.
  - Softplus derivatives vanish. Gradients become small.